data visualisation

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LOAD THE NECESSARY PACKAGES

```r
library("dplyr")
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library("lubridate")
```

```
##
## Attaching package: 'lubridate'

## The following objects are masked from 'package:base':
##
##     date, intersect, setdiff, union
```

```r
library("tidyr")
library("ggplot2")
```

SET THE WORKING DIRECTORY

```r
setwd("C:/Users/Admin/Desktop/DML")
```

LOAD THE WORKING DIRECTORY

```r
df<-read.csv("StudentsPerformance.csv")
```

Load required libraries

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats 1.0.1     v stringr 1.6.0
## v purrr   1.2.1     v tibble  3.3.1
## v readr   2.2.0
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

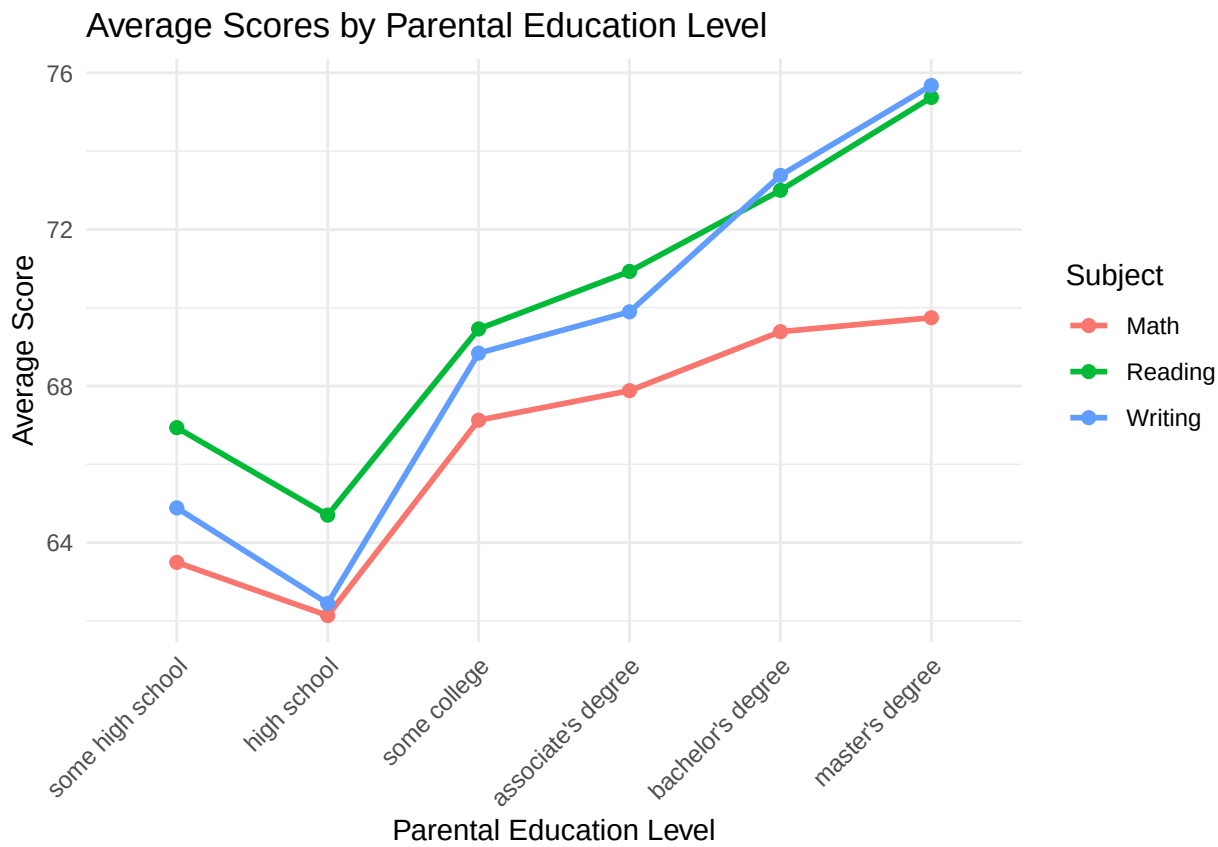
```r
library(ggplot2)
```

Question 1: Line graph - Average scores by parental education level

```r
q1_data <- df %>%
  pivot_longer(cols = c(math.score, reading.score, writing.score),
               names_to = "subject",
               values_to = "score") %>%
  group_by(parental.level.of.education, subject) %>%
  summarise(avg_score = mean(score), .groups = 'drop') %>%
  mutate(subject = case_when(
    subject == "math.score" ~ "Math",
    subject == "reading.score" ~ "Reading",
    subject == "writing.score" ~ "Writing"
  ))

# Reorder education levels
edu_order <- c("some high school", "high school", "some college",
               "associate's degree", "bachelor's degree", "master's degree")
q1_data$parental.level.of.education <- factor(q1_data$parental.level.of.education,
                                              levels = edu_order)

# Plot Q1
ggplot(q1_data, aes(x = parental.level.of.education, y = avg_score,
                    group = subject, color = subject)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Average Scores by Parental Education Level",
       x = "Parental Education Level",
```

```
       y = "Average Score",
       color = "Subject")
```

```
## Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use `linewidth` instead.
## This warning is displayed once per session.
## Call `lifecycle::last_lifecycle_warnings()` to see where this warning was
## generated.
```

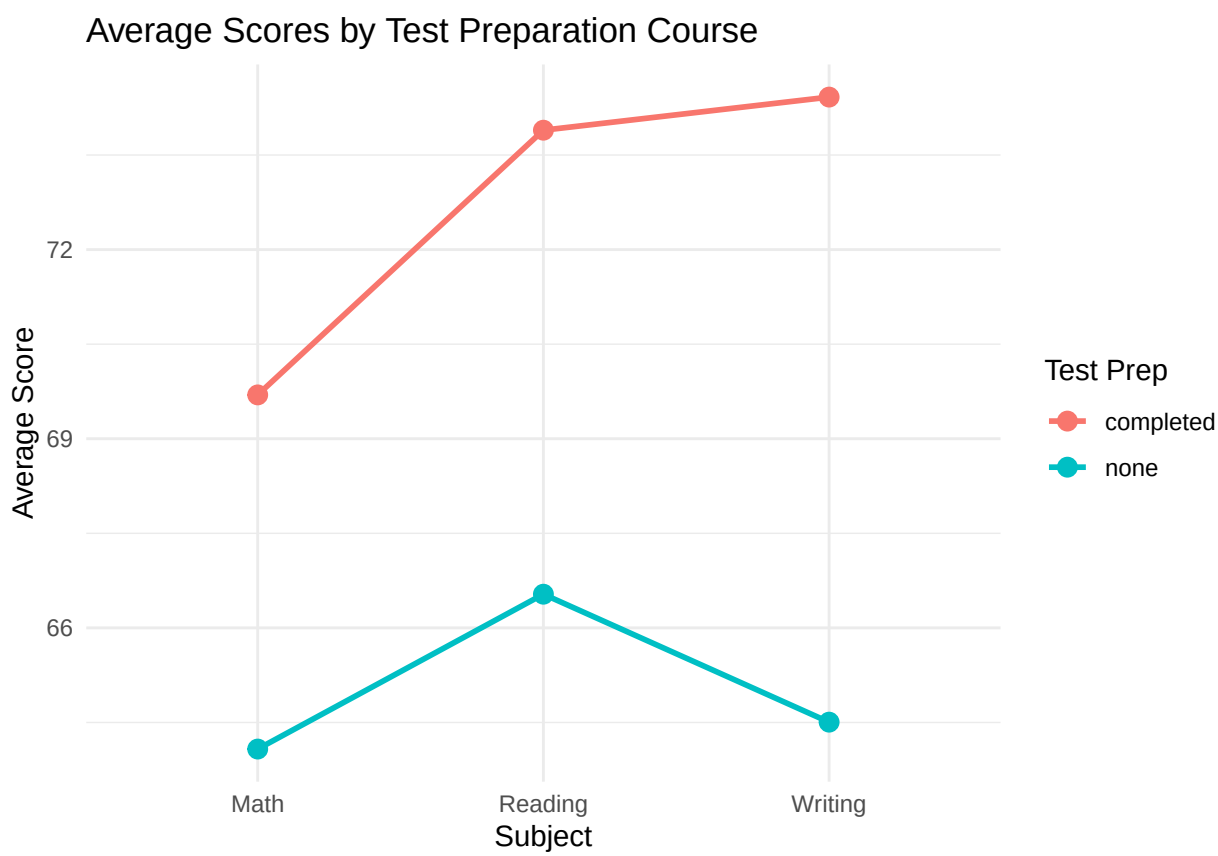



```
# Analysis: Which subject shows strongest improvement?
q1_improvement <- q1_data %>%
  group_by(subject) %>%
  summarise(improvement = max(avg_score) - min(avg_score)) %>%
  arrange(desc(improvement))
# Writing shows the strongest improvement
```

Question 2: Line graph - Test preparation comparison

```
q2_data <- df %>%
  pivot_longer(cols = c(math.score, reading.score, writing.score),
               names_to = "subject",
```

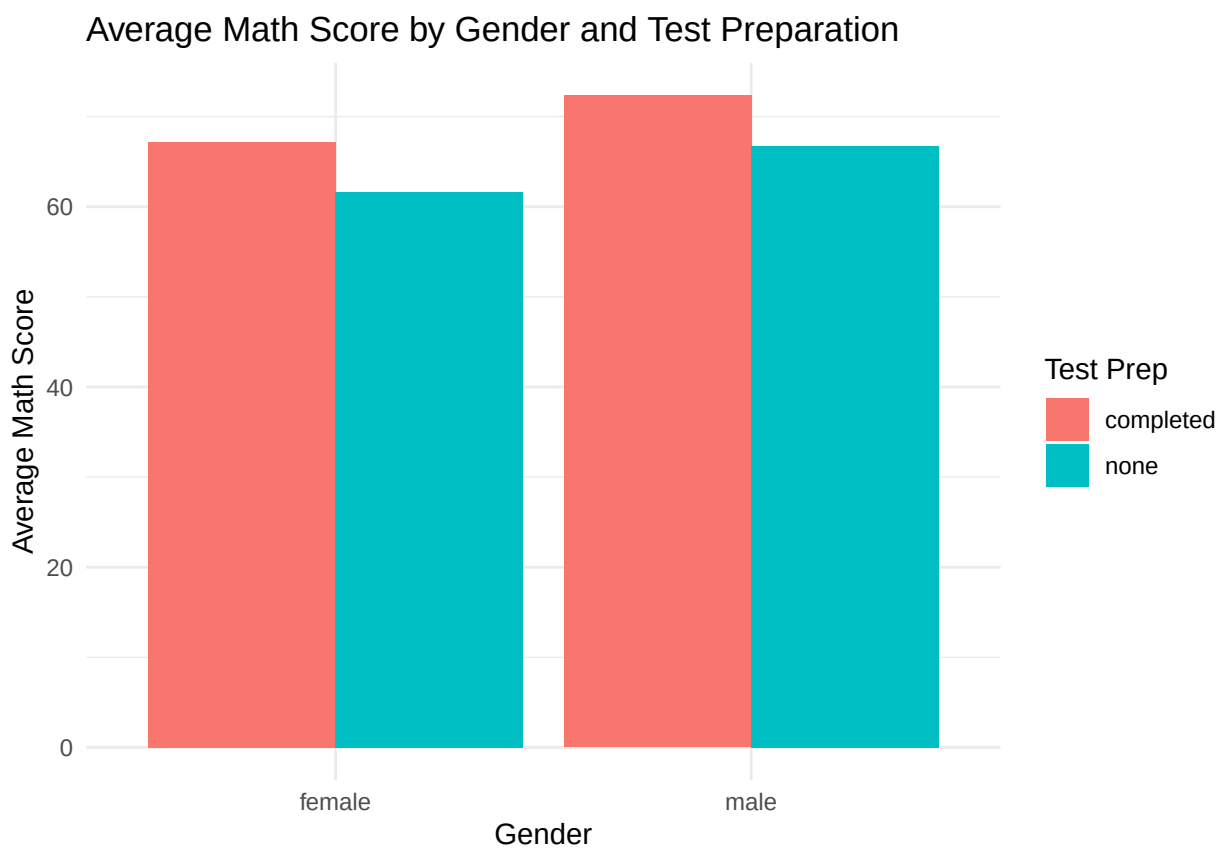
```
                values_to = "score") %>%
  group_by(test.preparation.course, subject) %>%
  summarise(avg_score = mean(score), .groups = 'drop') %>%
  mutate(subject = case_when(
    subject == "math.score" ~ "Math",
    subject == "reading.score" ~ "Reading",
    subject == "writing.score" ~ "Writing"
  ))

ggplot(q2_data, aes(x = subject, y = avg_score,
                    group = test.preparation.course,
                    color = test.preparation.course)) +
  geom_line(size = 1) +
  geom_point(size = 3) +
  theme_minimal() +
  labs(title = "Average Scores by Test Preparation Course",
       x = "Subject", y = "Average Score",
       color = "Test Prep")
```



Question 3: Grouped bar chart - Math score by gender and test prep

```
q3_data <- df %>%
  group_by(gender, test.preparation.course) %>%
  summarise(avg_math = mean(math.score), .groups = 'drop')

ggplot(q3_data, aes(x = gender, y = avg_math,
```

```r
                    fill = test.preparation.course)) +
  geom_col(position = "dodge") +
  theme_minimal() +
  labs(title = "Average Math Score by Gender and Test Preparation",
       x = "Gender", y = "Average Math Score",
       fill = "Test Prep")
```

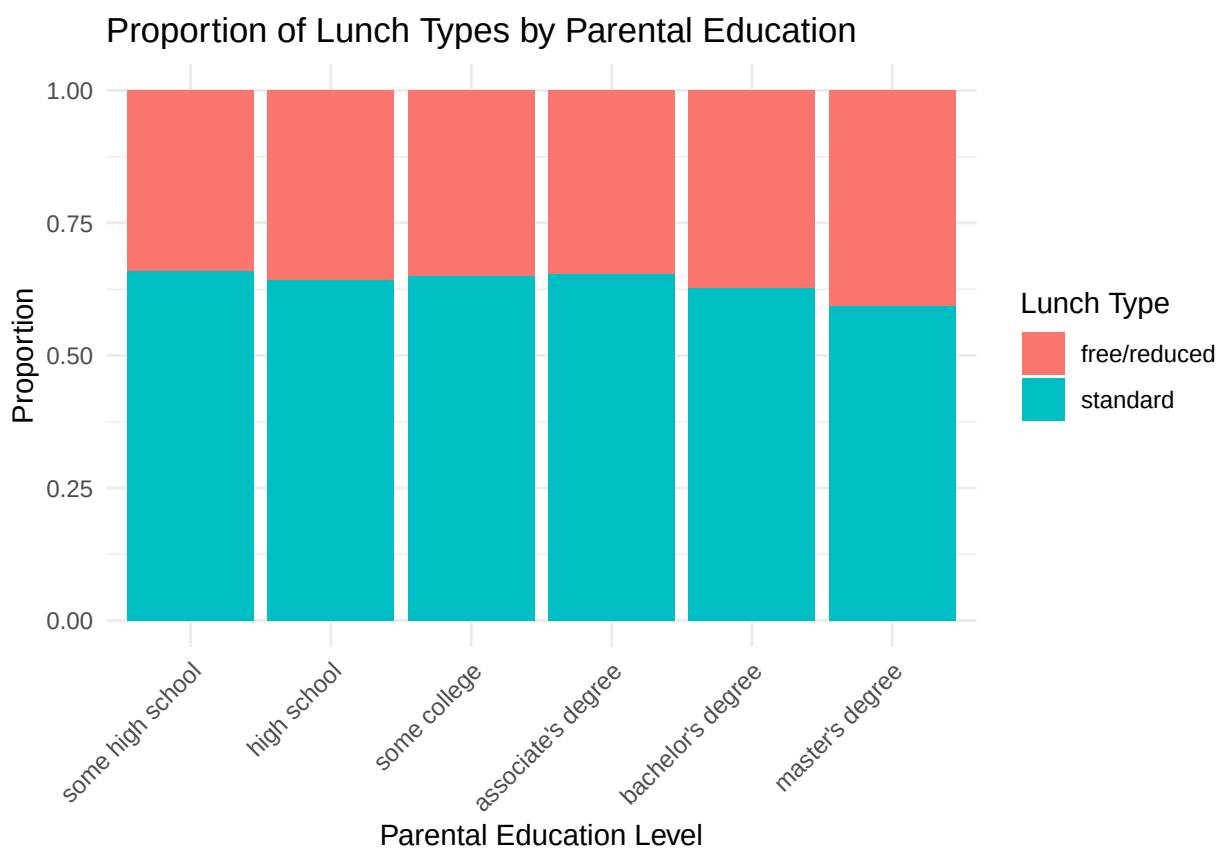



```r
# Highest average math score group
q3_data %>% arrange(desc(avg_math))  # Female with completed prep
```

```
## # A tibble: 4 x 3
##   gender test.preparation.course avg_math
##   <chr>  <chr>                      <dbl>
## 1 male   completed                   72.3
## 2 female completed                   67.2
## 3 male   none                        66.7
## 4 female none                        61.7
```

Question 4: Stacked bar chart - Students distribution by parental education and lunch

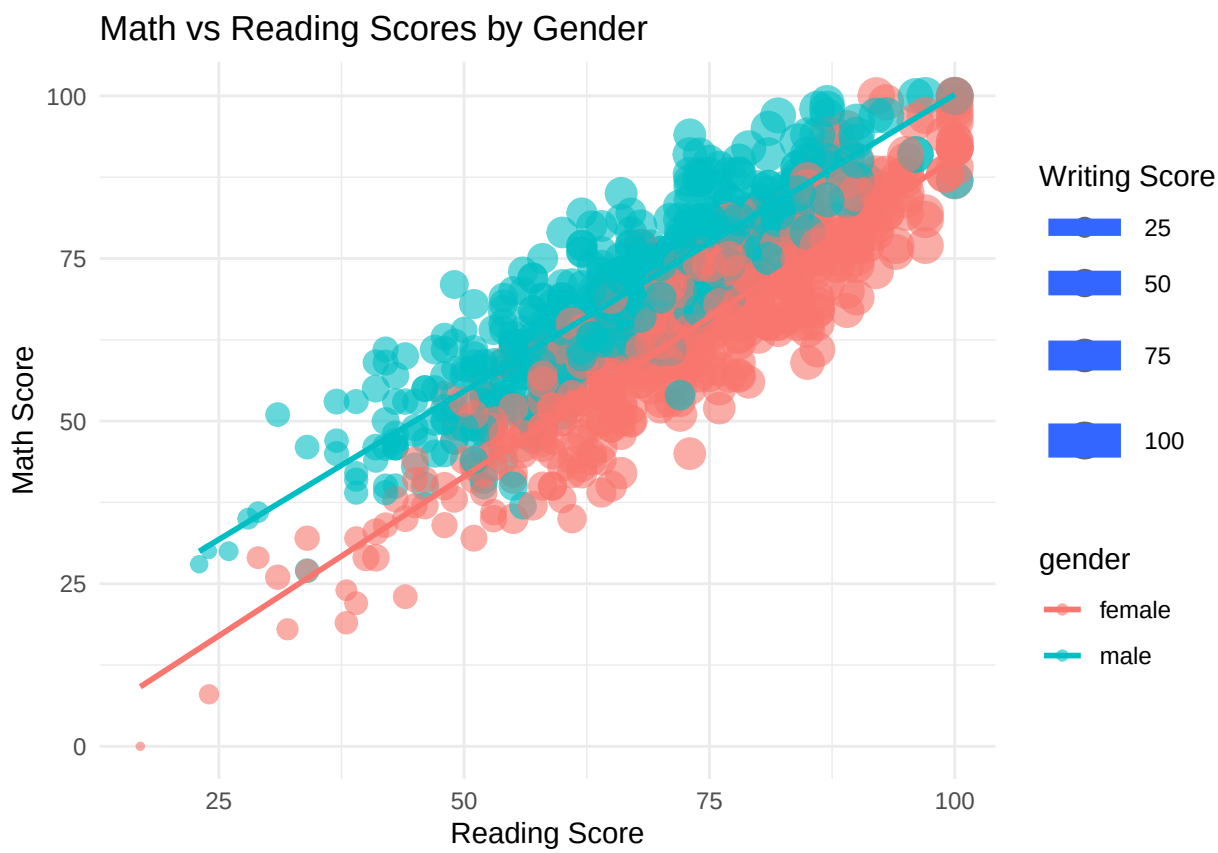
```r
q4_data <- df %>%
  group_by(parental.level.of.education, lunch) %>%
  summarise(count = n(), .groups = 'drop')

q4_data$parental.level.of.education <- factor(q4_data$parental.level.of.education,
                                              levels = edu_order)

ggplot(q4_data, aes(x = parental.level.of.education, y = count,
                    fill = lunch)) +
  geom_col(position = "fill") +
  theme_minimal() +
  theme(axis.text.x = element_text(angle = 45, hjust = 1)) +
  labs(title = "Proportion of Lunch Types by Parental Education",
       x = "Parental Education Level", y = "Proportion",
       fill = "Lunch Type")
```



Question 5: Scatter plot - Math vs Reading with gender color

```r
ggplot(df, aes(x = reading.score, y = math.score,
               color = gender, size = writing.score)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method = "lm", se = FALSE, aes(group = gender)) +
  theme_minimal() +
  labs(title = "Math vs Reading Scores by Gender",
       x = "Reading Score", y = "Math Score",
       size = "Writing Score")
```

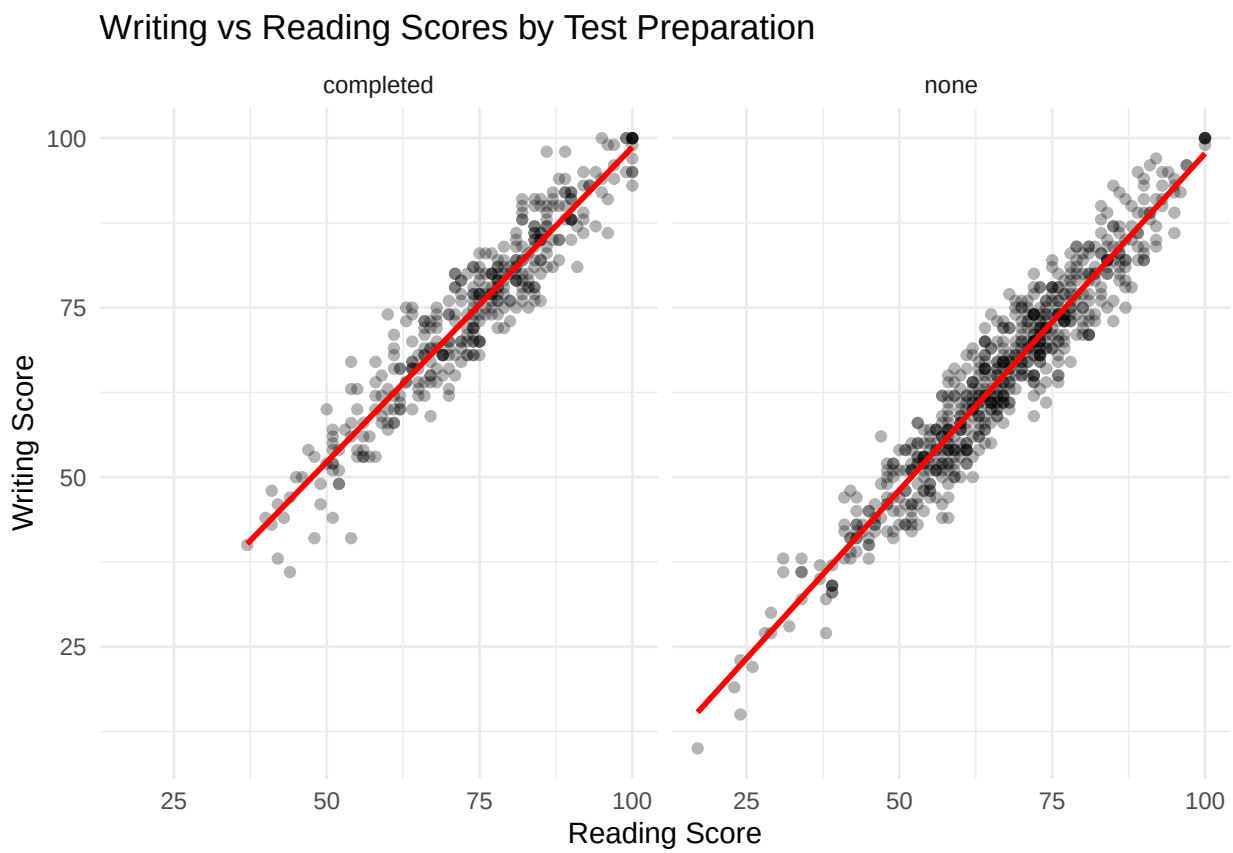
```
## `geom_smooth()` using formula = 'y ~ x'

## Warning: The following aesthetics were dropped during statistical transformation: size.
## i This can happen when ggplot fails to infer the correct grouping structure in
##   the data.
## i Did you forget to specify a `group` aesthetic or to convert a numerical
##   variable into a factor?
```



Question 6: Scatter plot - Writing vs Reading faceted by test prep

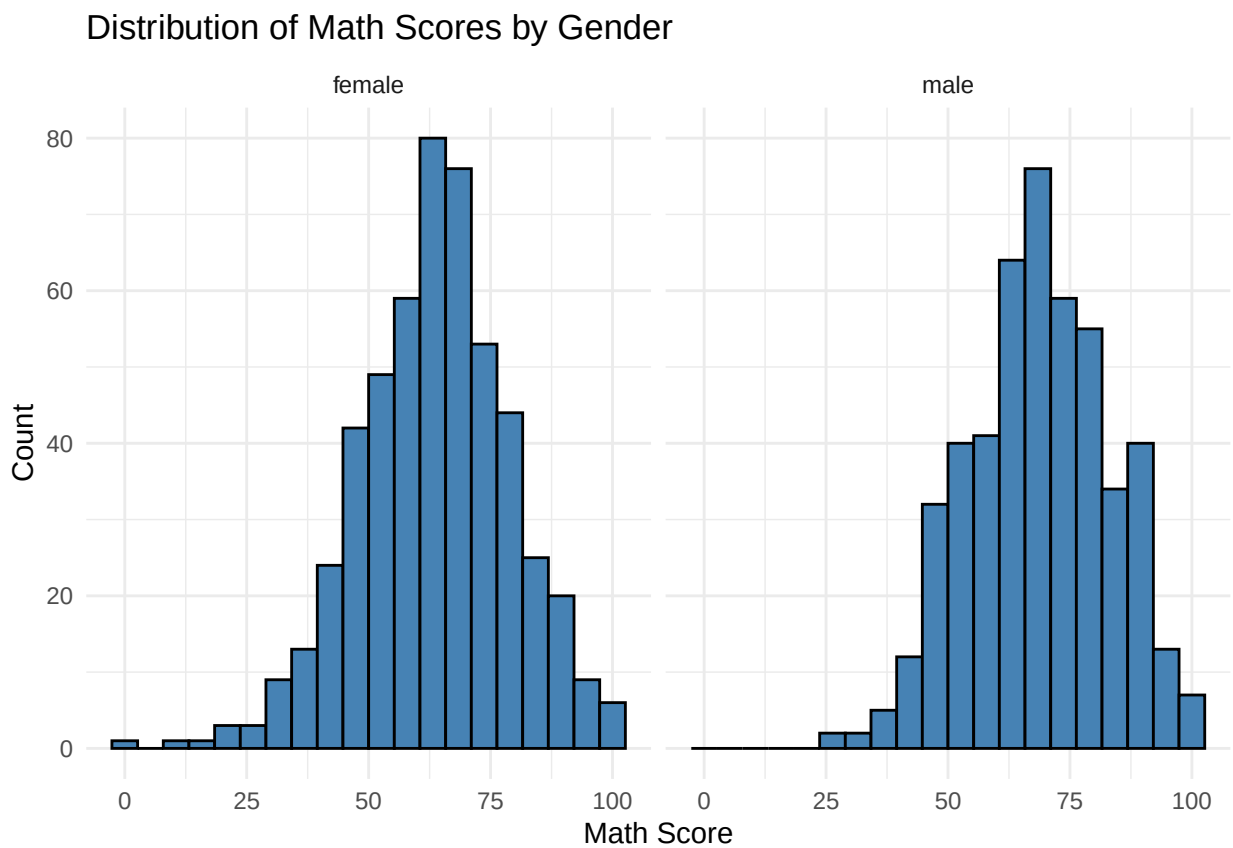
```r
ggplot(df, aes(x = reading.score, y = writing.score)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE, color = "red") +
  facet_wrap(~test.preparation.course) +
  theme_minimal() +
  labs(title = "Writing vs Reading Scores by Test Preparation",
       x = "Reading Score", y = "Writing Score")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



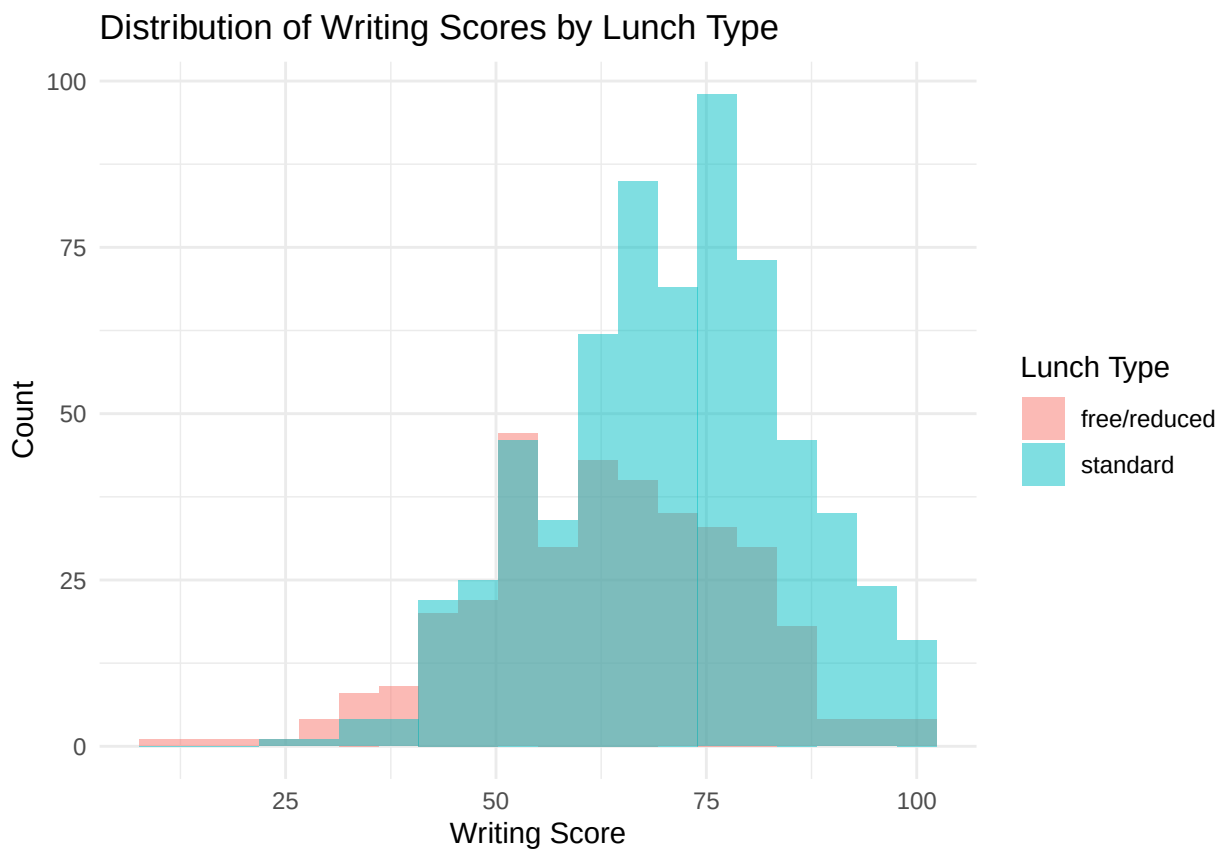
Question 7: Histogram - Math score distribution by gender

```
ggplot(df, aes(x = math.score)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "black") +
  facet_wrap(~gender) +
  theme_minimal() +
  labs(title = "Distribution of Math Scores by Gender",
       x = "Math Score", y = "Count")
```



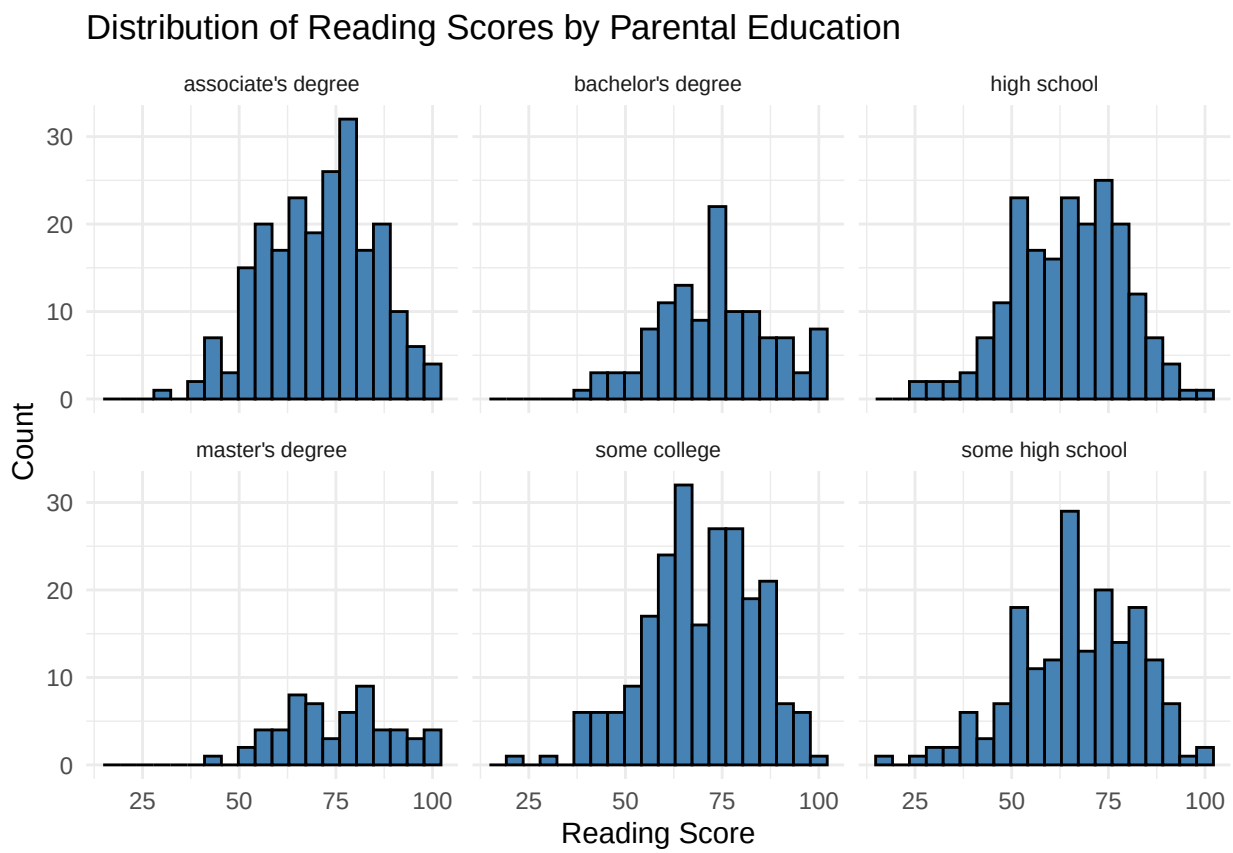
Question 8: Histogram - Writing scores by lunch type

```
ggplot(df, aes(x = writing.score, fill = lunch)) +
  geom_histogram(bins = 20, position = "identity", alpha = 0.5) +
  theme_minimal() +
  labs(title = "Distribution of Writing Scores by Lunch Type",
       x = "Writing Score", y = "Count",
       fill = "Lunch Type")
```



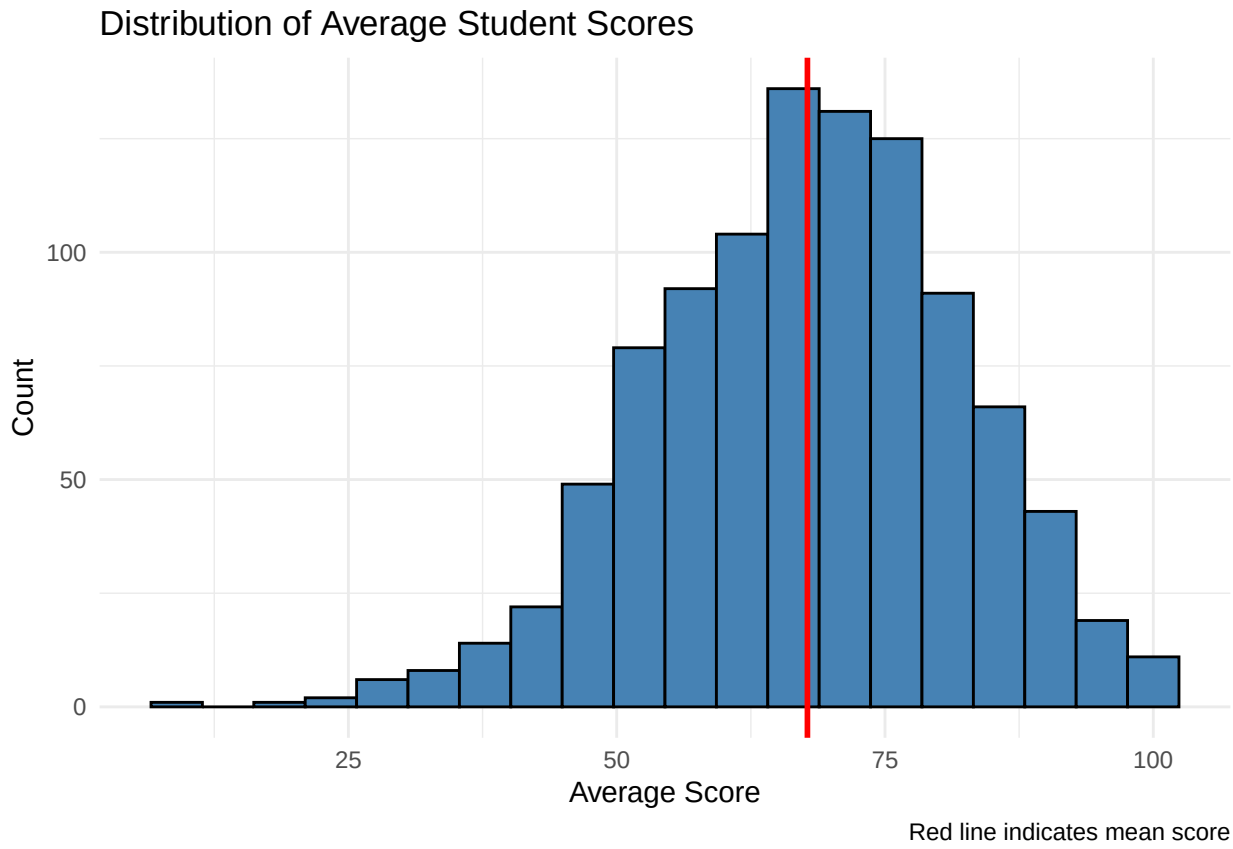
Question 9: Faceted histogram - Reading scores by parental education

```
ggplot(df, aes(x = reading.score)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "black") +
  facet_wrap(~parental.level.of.education) +
  theme_minimal() +
  theme(strip.text = element_text(size = 8)) +
  labs(title = "Distribution of Reading Scores by Parental Education",
       x = "Reading Score", y = "Count")
```



Question 10: Average score distribution

```
q10_data <- df %>%
  mutate(avg_score = (math.score + reading.score + writing.score) / 3)

mean_score <- mean(q10_data$avg_score)

ggplot(q10_data, aes(x = avg_score)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "black") +
  geom_vline(xintercept = mean_score, color = "red", size = 1) +
  theme_minimal() +
  labs(title = "Distribution of Average Student Scores",
       x = "Average Score", y = "Count",
       caption = "Red line indicates mean score")
```


Distribution of Average Student Scores
Count
100
50
0
25
50
75
100
Average Score
Red line indicates mean score